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Predicting Student Academic Performance from Collaborative Learning Behaviors Using Machine Learning and Explainable Insights

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This work examines whether patterns of collaborative learning can be used to anticipate university students' academic performance through machine learning. The study draws on responses from 690 students collected via a Google Forms questionnaire. The survey captures both basic demographic information and a range of collaboration-related behaviors, resulting in a dataset of 13 features, including age, how often and how long students collaborate, typical group size, perceived helpfulness, improvement in understanding, perceived effects on grades and stress, motivation level, study preference, study location, and overall perception of collaboration.

Before model development, the data were preprocessed by converting categorical responses into numeric form, and class imbalance was handled using an oversampling approach. Multiple supervised learning algorithms were then trained and tested using a held-out portion of the data. Among the evaluated models, ensemble-based methods produced the most consistent results, with the top model reaching an accuracy of 0.977 and a ROC-AUC close to 0.99, indicating strong predictive performance.

An examination of feature contributions suggests that the level of engagement in collaborative activities and students' perceived learning benefits are particularly influential in predicting outcomes. Taken together, these results indicate that even lightweight, survey-based indicators of collaboration can support practical predictive models, which may be useful for early identification of performance patterns and for guiding targeted academic support strategies.

1.1 INTRODUCTION

Collaborative learning is widely used in higher education because it promotes discussion, shared problem solving, and knowledge construction through peer interaction

across different disciplines and learning settings. Prior research consistently links peer engagement with academic achievement. For instance, Driessen et al. [1] reported significant performance improvements from structured group work, while Chen et al. [4] found measurable peer effects on GPA, suggesting that academic outcomes are partly influenced by peer environments.

Collaboration is not only about how frequently students interact; it also reflects a structured social process that can be analyzed through interaction patterns and connectivity. Crossette et al. [2] showed that students' positions within learning networks relate to performance, and Swain [3] demonstrated that WiFi-based co-location patterns correlate with academic outcomes. Similarly, Castillo et al. [14] emphasized the role of relational factors such as trust and informal communication, while Calunsag and Calunsag [5] noted that digital networking supports learning when interactions remain academically focused.

Studies focusing on underlying mechanisms explain why collaboration is beneficial in some contexts but not others. Zhang et al. [8] linked peer interaction to achievement through self-efficacy, and Premo et al. [9] highlighted motivational benefits from peer experiences. However, interaction quality is critical. Anghel [12] and Premo et al. [11] showed that task-focused collaboration is more effective than purely social interaction. In addition, collaborative effects vary across contexts, as demonstrated by Dittmann et al. [10], Zhou et al. [13], Hu et al. [16], and Paabo et al. [17], while Linn et al. [7] and Oroka et al. [15] further support the academic value of peer-based activities.

Despite strong explanatory findings, predictive applications remain limited, with few studies using collaboration indicators for early academic risk prediction. This study addresses that gap by treating collaboration variables as model features and evaluating multiple machine learning classifiers within a reproducible framework.

The main contributions of this study are as follows:

- Built a reproducible ML pipeline to predict academic performance from collaboration indicators.
- 690 data were collected from university students in Bangladesh using structured survey forms to ensure broad representation.
- Developed collaboration-related features along with appropriate preprocessing steps, including encoding, scaling, and handling class imbalance.
- Evaluated multiple machine learning models (LR, SVM, KNN, AdaBoost, Gradient Boosting, Extra Trees, Random Forest, and XGBoost) using standard performance metrics.
- Produced analysis outputs such as confusion matrices, ROC–AUC curves, feature importance, and SHAP-based interpretations for reporting.
- Identified influential collaboration factors to support instructional decisions and early support.

1.2 LITERATURE REVIEW

Recent studies (2020–2026) consistently show that collaborative learning and peer interaction are closely linked to academic performance across different disciplines and learning environments. For example, Driessen et al. [1] provided strong quantitative evidence through a meta-analysis of group work in life science courses, reporting an average improvement of about one standard deviation. Supporting this, Chen et al. [4] used a quasi-experimental design based on roommate assignments to identify clear peer effects on GPA, while Abio et al. [6] reported similar findings in economics. At the same time, the medium of interaction matters. Calunsag and Calunsag [5] observed that social media can support learning, but its effect depends on whether it is used for academic purposes or becomes a source of distraction.

Beyond outcome-based studies, recent work has increasingly focused on behavioral patterns to understand how interaction structure influences performance. Crossette et al. [2] applied social network analysis during pandemic-era courses and found that students' positions within networks are associated with homework performance. Similarly, Swain [3] used WiFi-based co-location data to show that proximity-based interaction patterns correlate with academic results ($r = 0.24$) and can sometimes outperform models relying only on individual-level features. Castillo et al. [14] further emphasized that group outcomes depend not just on participation levels but also on the underlying social structure of interactions.

Mechanism-oriented studies help explain why collaboration is effective in some cases but not in others. Zhang et al. [8] demonstrated that peer interaction contributes to achievement partly through self-efficacy, indicating that students' confidence plays a mediating role. Premo et al. [9] also highlighted the motivational benefits of peer engagement in science education. However, the quality of interaction is critical. Anghel [12] found that meaningful, task-focused collaboration predicts performance more reliably than simple interaction frequency, and Premo et al. [11] showed that active contribution improves outcomes, whereas purely social interaction may be less beneficial.

Context and instructional design also influence collaborative outcomes. Dittmann et al. [10] showed that collaboration may benefit students differently depending on their socio-cultural background. Zhou et al. [13] provided experimental evidence that study groups improve performance, particularly for less-prepared students, although they may introduce time-management challenges. In practical learning settings, Hu et al. [16] found that structured, technology-supported teamwork enhances self-directed learning, while Paabo et al. [17] highlighted the role of organized peer-learning programs in supporting academic progress.

1.2.1 Comparison with Prior Predictive Work

Although there is a substantial body of explanatory research, the use of collaborative learning variables for predictive modeling of academic performance remains relatively limited. Most existing studies consider collaboration as a factor influencing outcomes rather than as a set of predictive features. Table 1.1 presents a comparison between the present study and selected related works.

Table 1.1 Comparison with prior academic performance prediction studies

Study	Data Type	ML Models Used	Best Acc.	Key Limitation
Swain et al. [3]	WiFi collocation logs	Statistical/correlation-based analysis	N/A	No direct ML classifier; uses indirect behavioral proxy
Hu et al. [16]	Classroom learning system data	Experimental/statistical analysis	N/A	No predictive modeling; lacks ML-based evaluation
Zhang et al. [8]	Student interaction data	Mediation/behavioral analysis	N/A	Focuses on relationships; no performance prediction
Present Study	Collaboration survey (N=690)	8 classifiers + SHAP	0.977	Self-reported; single institution

Table 1.1 shows that most prior studies mainly focus on explaining how collaborative learning behaviors relate to academic performance, rather than building predictive models. For example, studies such as Swain et al. [3], Hu et al. [16], and Zhang et al. [8] analyze collaboration patterns, learning environments, and student interactions using statistical or behavioral approaches. While these works provide useful insights into peer learning and academic outcomes, they do not directly apply machine learning for performance prediction.

In contrast, the present study takes a predictive approach by using collaboration-related variables as input features within a supervised machine learning framework. Multiple classifiers are evaluated under a consistent preprocessing pipeline, allowing a more systematic comparison of model performance. In addition, SHAP-based interpretability is used to examine how individual features influence predictions, addressing a limitation in earlier explanatory studies.

Although direct numerical comparison across studies is difficult due to differences in data and evaluation methods, the proposed approach demonstrates strong predictive performance, achieving an accuracy of approximately 0.977. This indicates that collaboration-based behavioral features can be effectively used for academic performance prediction.

Finally, this study introduces a dataset collected from Bangladeshi university students, a context that is still underrepresented in existing research. Despite limitations such as reliance on self-reported data and a single-institution sample, the results highlight the practical value of combining collaborative learning insights with machine learning for academic performance analysis.

1.3 METHODOLOGY

This study proposes a supervised machine learning pipeline to predict students' academic performance from collaboration-related survey indicators. The methodological

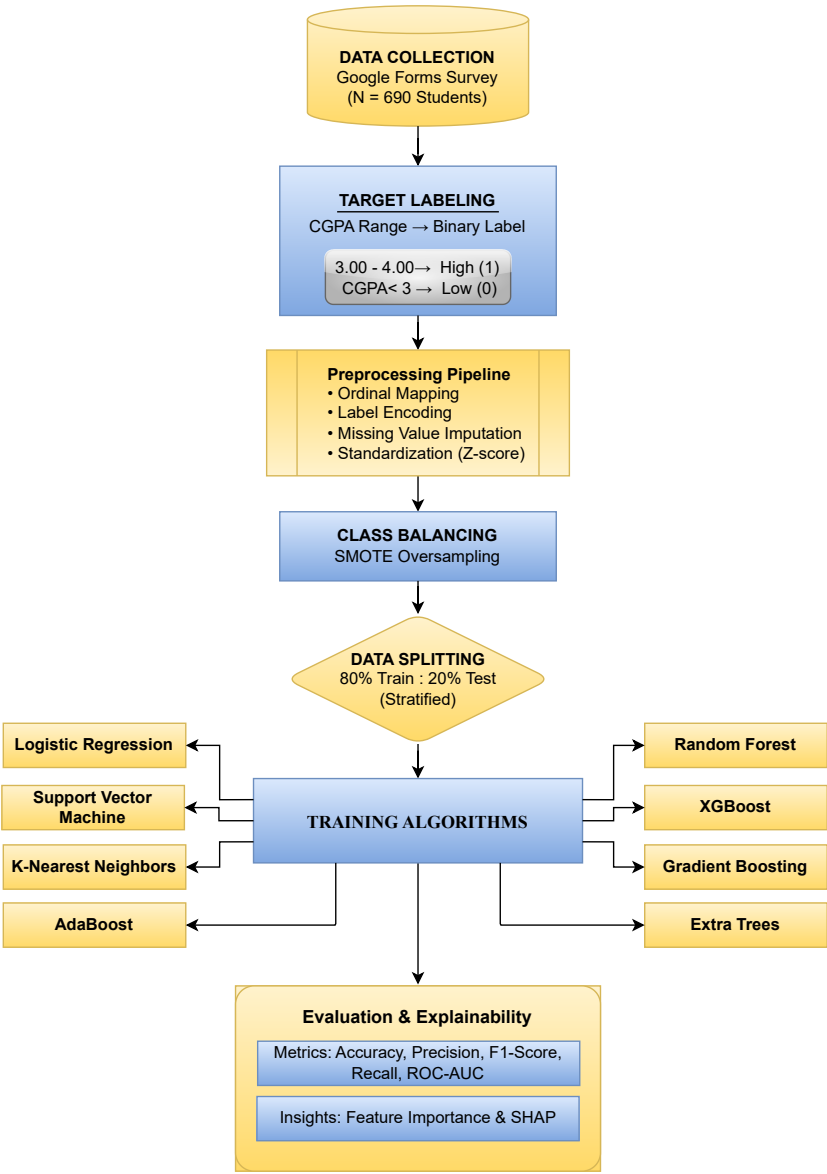


Figure 1.1 Methodology Diagram.

design relies on self-reported collaborative learning variables and transforms them into structured predictive features for benchmarking multiple classifiers.

1.3.1 Dataset Collection

A cross-sectional questionnaire survey was administered using Google Forms to collect self-reported responses from university students (N = 690). The questionnaire captured demographic information and collaboration-related indicators describing students' learning practices and perceived effects of group study. The collected dataset

contained 13 variables (columns), including age, collaboration frequency and duration, group size, perceived helpfulness, understanding improvement, perceived grade and stress effects, motivation score, solo-vs-group preference, study location, and overall perceived effect.

1.3.2 Dataset Annotation (Binary Target Construction)

Academic performance was recorded as an ordinal category (CGPA range). For early-warning prediction, the target was transformed into a binary label using a transparent proxy rule implemented in the workflow. Let y_i denote the binary label for student i , defined as:

$$y_i = \begin{cases} 1, & \text{if CGPA_range}_i \in \{3.00\text{--}3.50, 3.50\text{--}4.00\} \\ 0, & \text{otherwise} \end{cases} \quad (1.1)$$

Table 1.2 Proxy labeling rule for binary academic performance (*PerformanceLabel*).

CGPA range category	Label
3.50–4.00 (and equivalent variants)	1 (High/Good performance)
3.00–3.50 (and equivalent variants)	1 (High/Good performance)
Other CGPA ranges	0 (Low/Needs improvement)

1.3.3 Feature Selection Rationale

The 13 features used in this study were chosen with reference to established findings in collaborative learning and their relevance to academic performance. Age was included as a control variable, as earlier studies associate it with study behavior and self-regulation. Collaboration frequency and duration represent how often and how long students engage with peers, both of which are linked to knowledge development and academic outcomes [1], [12]. Group size captures the structural aspect of collaboration, with prior work suggesting that smaller groups tend to enable more effective interaction [14].

Perceived helpfulness and understanding improvement reflect students' own evaluation of whether collaboration supports their learning. These perceptions are closely related to self-efficacy and academic performance [8]. In a similar way, perceived grade effect and stress effect indicate how students believe collaboration influences their results and wellbeing, which can shape their level of engagement [9], [11]. Motivation score represents the degree of willingness to participate in collaborative activities and is commonly associated with improved outcomes [10].

The solo-versus-group preference variable indicates whether a student tends to work independently or in groups, a factor that can influence how effective collaboration becomes [10], [13]. Study location represents the primary setting where collaboration occurs, such as at home, in a library, or online, and serves as an indicator of the learning environment. The overall perceived effect combines students' general impressions of collaboration into a single measure.

Taken together, these features provide a structured representation of collaborative learning from multiple perspectives, including behavioral, structural, motivational, and perceptual aspects.

1.3.4 Preprocessing and Feature Engineering

Survey responses were transformed into numeric inputs using predefined ordinal mappings and label encoding. Ordinal response categories (e.g., frequency levels, duration groups, Likert-type agreement levels, and location categories) were converted into numeric scales using a mapping dictionary, and any remaining text fields were label-encoded. All features were coerced into numeric form and missing values were handled to ensure a complete training matrix. Feature standardization (z-score normalization) was applied to scale-sensitive models. To address class imbalance, SMOTE-based

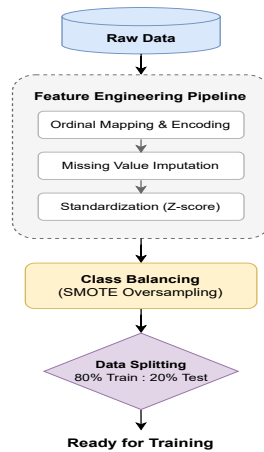


Figure 1.2 Preprocessing steps.

oversampling was used when required. Finally, the dataset was split into training and testing sets using an 80:20 stratified split with a fixed random seed (*random_state* = 42).

1.3.5 Class Balancing with SMOTE

The original dataset exhibited a class imbalance, with a substantially higher proportion of students in the High/Good performance category ($\text{CGPA} \geq 3.00$) relative to the Low performance category. This imbalance, if left unaddressed, can cause classifiers to become biased toward the majority class, resulting in inflated accuracy but poor recall for the minority class a particularly critical limitation in early-warning prediction scenarios where correctly identifying at-risk students is paramount. To address this, Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training data prior to model fitting. SMOTE generates synthetic samples for the minority class by interpolating between existing minority-class instances in the feature space, rather than simply duplicating existing samples. This approach increases the representation of the minority class while reducing the risk of overfitting that can

occur with simple duplication of samples. In this study, SMOTE was applied only to the training data after performing an 80:20 stratified split. As a result, the test set remained unchanged and contained no synthetic samples. This helps maintain a fair evaluation setup and avoids any potential data leakage.

Table 1.3 Feature set used for prediction.

Feature	Description
age	Student age (numeric or mapped group)
frequency	Collaboration frequency (ordinal mapped)
duration	Collaboration duration (ordinal mapped)
groupsize	Typical group size
helpfulness	Perceived helpfulness (agreement scale)
understanding	Perceived understanding improvement (agreement scale)
gradeeffect	Perceived grade impact (agreement scale)
stresseffect	Perceived stress impact (mapped scale)
motivationscore	Motivation level / score (mapped scale)
solovsgroup	Solo vs. group preference (mapped)
location	Primary collaboration location (label encoded)
overalleffect	Overall perceived effect (agreement scale)

Table 1.4 Ordinal mappings used to transform survey responses into numeric features.

Variable group	Categories → numeric mapping
Frequency	Never→0, Rarely→1, Monthly→2, Weekly→3, Daily→4
Duration	<30 mins→0, 30–60 mins→1, 1–2 hours→2, >2 hours→3
Agreement scale	Strongly Disagree→1, Disagree→2, Neutral→3, Agree→4, Strongly Agree→5
Yes/No items	No→0, Yes→1, Maybe→0.5
Stress/Motivation	None→0, Low→1, Medium→2, High→3
Location	Library→1, Online→2, Classroom→3, Cafe→4, Home→5 (label encoded)

1.3.6 Experimental Setup and Models

Eight supervised classifiers were evaluated using the same preprocessing pipeline, including Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), AdaBoost, Gradient Boosting, Extra Trees, Random Forest, and XGBoost. All models were trained on the training set and tested on a separate held-out set to ensure a fair and consistent comparison.

1.3.7 Evaluation Metrics

Model performance was evaluated using standard evaluation metrics based on the confusion matrix, where TP , FP , TN , and FN represent true positives, false positives, true negatives, and false negatives, respectively, with respect to the positive

class (High/Good performance).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1.2)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1.3)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (1.4)$$

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (1.5)$$

$$\text{TPR} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN} \quad (1.6)$$

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}) d(\text{FPR}) \quad (1.7)$$

1.4 RESULTS AND DISCUSSION

This section presents the evaluation of the supervised machine learning models used to predict academic performance based on collaboration-related survey features. The results are interpreted using standard classification metrics, confusion matrix patterns, and model explainability analysis.

1.4.1 Experimental Setup

All experiments were carried out in Google Colab using Python, along with commonly used machine learning libraries such as Scikit-learn and XGBoost. The dataset was collected through a Google Form survey and includes variables related to collaborative learning, including *age*, *frequency*, *duration*, *group size*, *perceived helpfulness*, *perceived understanding*, *grade effect*, *stress effect*, *motivation score*, *solo-vs-group preference*, *study location*, and *overall perceived effect*.

The academic outcome variable *cgparange* was transformed into a binary target using a proxy rule, where higher CGPA ranges were mapped to the positive class (High/Good performance). An 80:20 stratified train–test split with a fixed random seed (`random_state = 42`) was used.

Ordinal survey categories were mapped into numeric scales, and remaining categorical variables were label-encoded. Feature standardization (`StandardScaler`) was applied for scale-sensitive models, and SMOTE was applied to the training set to address class imbalance.

Eight classifiers were benchmarked: Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), AdaBoost, Gradient Boosting, Extra Trees, Random Forest, and XGBoost.

1.4.2 Hyperparameter Configuration

Table 1.5 Hyperparameter configuration for evaluated machine learning models.

Model	Hyperparameter settings
Logistic Regression	Default config (Scikit-learn); binary classification
SVM	<code>probability=True</code> , <code>random_state=42</code>
KNN	Default configuration (Scikit-learn)
AdaBoost	Default config, <code>random_state=42</code>
Gradient Boosting	Default config, <code>random_state=42</code>
Extra Trees	<code>n_estimators=200</code> , <code>random_state=42</code>
Random Forest	<code>n_estimators=200</code> , <code>random_state=42</code>
XGBoost	<code>eval_metric=logloss</code> , <code>use_label_encoder=False</code>

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1.4.3 Quantitative Model Evaluation

Table 1.6 Performance comparison of machine learning models on the test set.

Model	Acc	Prec	Rec	F1	AUC
Logistic Regression	0.928	0.874	1.000	0.933	0.964
KNN	0.932	0.881	1.000	0.937	0.979
AdaBoost	0.959	0.925	1.000	0.961	0.983
Gradient Boosting	0.977	0.957	1.000	0.978	0.990
SVM	0.977	0.957	1.000	0.978	0.982
Random Forest	0.977	0.957	1.000	0.978	0.993
XGBoost	0.977	0.957	1.000	0.978	0.992
Extra Trees	0.977	0.957	1.000	0.978	0.993

Table 1.6 reports the held-out test-set performance of all models. The results indicate that tree-based ensembles and boosting methods achieve the strongest overall performance, while the linear baseline performs comparatively lower, suggesting that collaboration-performance relationships are not purely linear.

1.4.4 Confusion Matrix and ROC Analysis

Figure 1.3 presents confusion matrices for all evaluated models on the test set. Figure 1.4 presents ROC curves comparing all models, with AUC values shown in the legend.

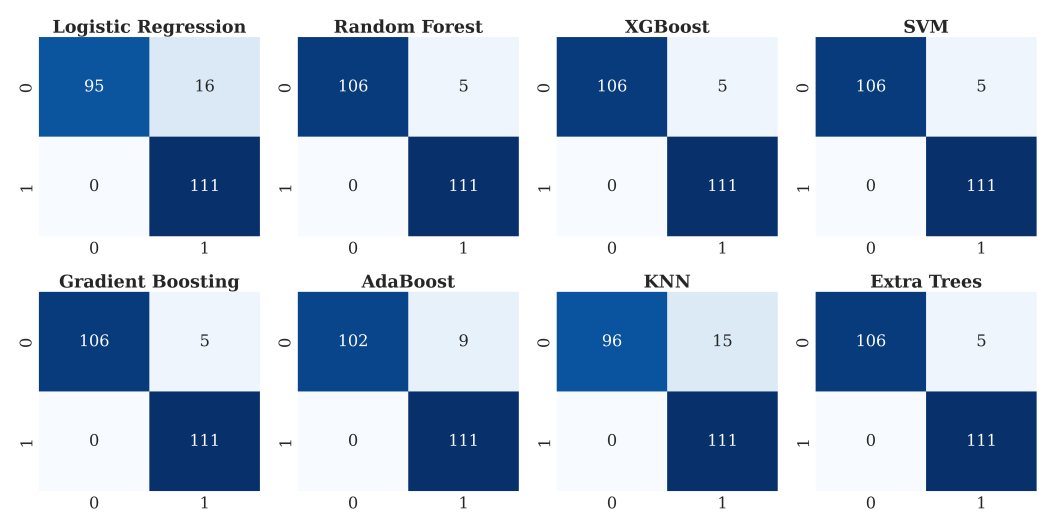


Figure 1.3 Confusion matrices of the evaluated classifiers on the test set.

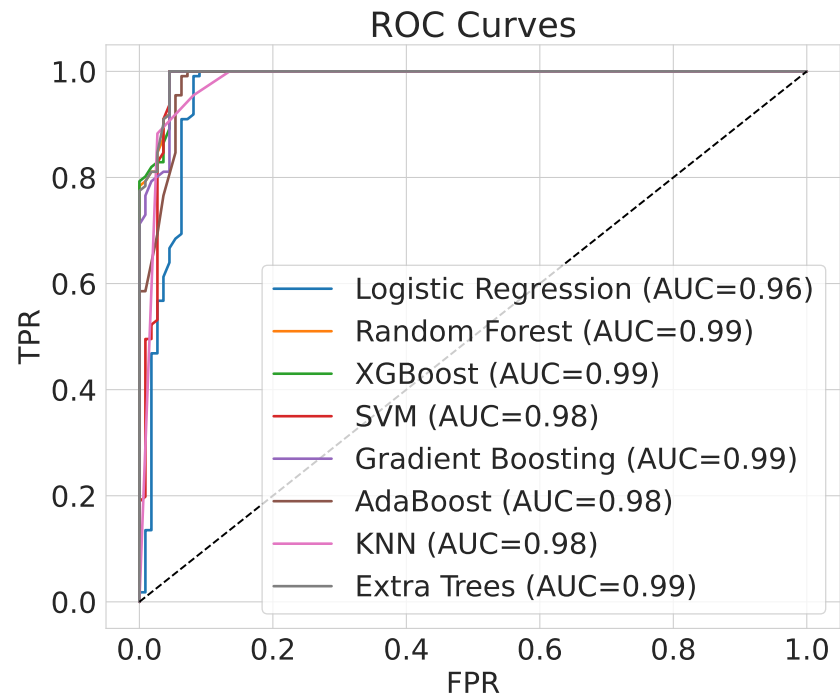


Figure 1.4 ROC curves of the evaluated models with AUC values on the held-out test set.

1.4.5 Feature Importance and Interpretability

To identify influential collaboration indicators, model interpretation was performed using a tree-based model. Figure 1.5 shows the global feature-importance ranking, and

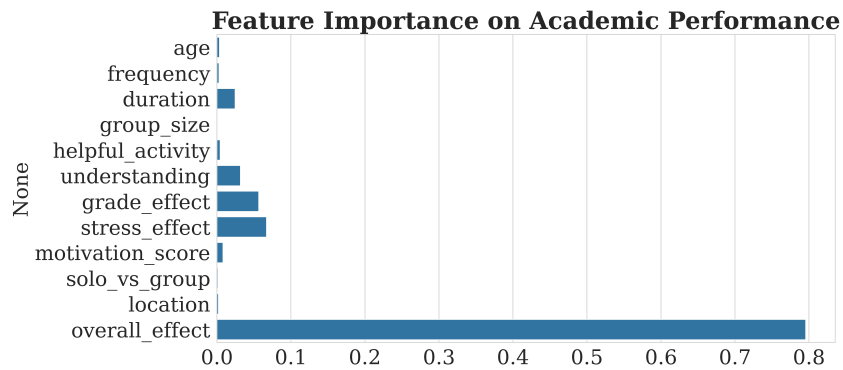


Figure 1.5 Global feature importance derived from the trained tree-based model.

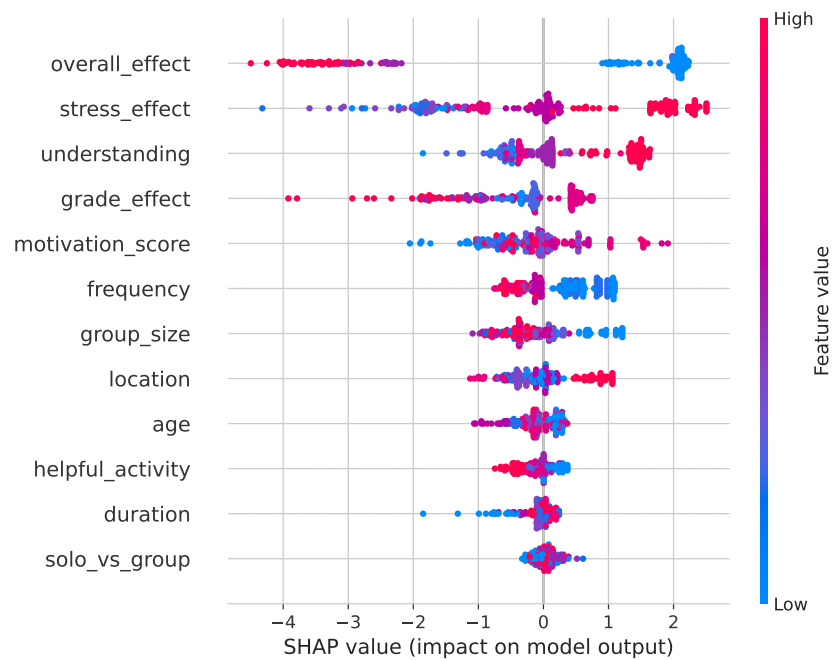


Figure 1.6 SHAP summary plot illustrating the impact of feature values on model output.

Figure 1.6 shows the SHAP summary plot illustrating both magnitude and direction of feature contributions.

1.4.6 Discussion

Overall, ensemble and other non-linear models achieved strong performance, indicating that academic performance differences are captured through a combination of collaboration behaviors and perceived learning effects. The interpretability outputs suggest that sustained collaboration engagement and its perceived academic value jointly contribute to model decisions. These results support the use of collaboration-related survey indicators as a practical basis for early academic monitoring, while emphasizing that predictions should be treated as decision-support signals rather than causal explanations.

1.5 CONCLUSION AND FUTURE WORK

This study shows that collaborative learning survey data can be used to predict academic performance effectively using supervised machine learning models. After preprocessing steps such as encoding, scaling, and applying SMOTE to handle class imbalance, and evaluating with an 80:20 stratified split, ensemble models (e.g., Random Forest and XGBoost) achieved strong performance (accuracy ≈ 0.977 and ROC-AUC close to 0.99). These results indicate that collaboration behavior and perceived learning impact provide useful predictive signals.

Several directions remain for future work:

- Validation on larger, multi-institution datasets: The current dataset comes from a single university in Bangladesh. Future studies should test the model on data from multiple institutions and different contexts to examine generalizability.
- Multi-class and regression-based targets: The binary CGPA setup simplifies performance differences. Extending to multi-class categories or continuous prediction could capture more detailed variation.
- Cross-validation and calibration: The current evaluation relies on a single 80:20 split. Using repeated k-fold cross-validation and probability calibration could improve reliability.
- Additional behavioral and academic features: Including LMS activity, attendance, assignment patterns, and long-term academic records may improve prediction and reduce reliance on self-reported data.
- Alternative methods for class imbalance: While SMOTE worked well, other approaches such as class weighting or ensemble-based methods could be explored for comparison.
- External dataset validation: The absence of external data is a limitation. Future work should include independent datasets to test generalizability.
- Exploring causal relationships: This study focuses on prediction. Future research could use causal methods or interventions to better understand underlying relationships.

- Deployment and fairness evaluation: Testing the model in real classroom settings, along with examining fairness and ethical considerations, would be an important next step.

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